

ENEL700 Tutorial 6

1. What is pulse amplitude modulation (PAM)? What is the difference between PAM and AM?

PAM is a form of signal modulation where the message information is encoded in the amplitude of a series of signal pulses. AM is an analogue modulation method where the amplitude of the carrier signal is varied according to the message signal.

2. What is the main difference between pulse width modulation (PWM) and pulse position modulation (PPM)? State one application of PWM.

In PWM, the pulse duration is varied according to the information signal, while in PPM, the pulse position is changed according to the amplitude of the information signal.

Common applications of PWM: DC/Servo motor speed control, brightness control of electric lighting, control charging current of batteries, data transmission, etc.

3. What are the two ways to generate pulse code modulation?

One is to use a sample and hold circuit, followed by an ADC to convert the analogue signal into a sequence of binary words, and then a shift register to convert the parallel binary words into serial form. The other is to use a delta modulator.

4. What is companding? Explain how companding is adopted in PCM.

Companding refers to the process of compressing the analogue signal at the transmitter and expanding the signal at the receiver. It is required in PCM to improve the SQNR, especially when input signal has a small amplitude.

At transmitting end, the analogue signal is fed to compressor circuit before feeding to a uniform quantizer. The dynamic range of the signal is decreased such that lower-level signals are emphasized and higher-level signals are deemphasized.

At receiving end, the recovered signal is fed to an expander circuit that does the opposite, deemphasizing lower-level signals and emphasizing higher-level signals, thereby returning the transmitted signal to its original condition.

5. Explain the advantages and disadvantages of a Delta Modulator.

Advantages include improved SQNR, reduced transmission bandwidth requirement. Disadvantages include granular and slope overload noises.